Materials Today: Proceedings 66 (2022) 2405–2410

Contents lists available at ScienceDirect

Materials Today: Proceedings

journal homepage: www.elsevier.com/locate/matpr

Simulation of wire arc additive manufacturing to find out the optimal path planning strategy

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ARTICLE INFO

Article history:
Available online 28 June 2022

Keywords:
Wire arc additive manufacturing
Residual stress
Scanning path-planning
Distortion
FEA

ABSTRACT

Wire Arc Additive Manufacturing is a metal 3-D printing technology that is a highly versatile and rapidly growing means of producing complex components by depositing metal layer-by-layer, melting them with an electric arc. Predicting the residual stress induced is of cardinal importance as components need to be analyzed during the design phase with considerations given to residual stresses to be effective. One of the critical parameters determining the residual stress of the structure in the process is the deposition pattern. In this study, finite element method in Simufact Welding 6.0 is employed to compare the different path-planning deposition strategies of the Wire Arc Additive Manufacturing process to understand how the dynamic temperature evolution can cause notable differences in distortion and residual stresses. The results show that the strategy that has the least chance of non-uniform cooling will have the least residual stresses, and the paths in which fewer longitudinal beads are exposed to the edges will tend to distort the least.
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1. Introduction

Additive manufacturing (AM) is a sort of non-conventional manufacturing in which solid objects are created by depositing constituent materials layer by layer [1]. Several AM technologies for free-form manufacturing of various materials have been developed over the last two decades. To ensure successful fabrication, each of these processes has its own set of process parameters that must be adjusted.

Wire Arc Additive Manufacturing (WAAM) is a Direct Energy Deposition (DED) additive manufacturing technique in which a continuously fed filler wire is melted using heat sources that stem from the nozzle. The droplets are deposited as metal layers, one over the other. This is much like the traditional welding process [2]. The main process parameters that influence the quality of the finished product in a WAAM process are the beam power, beam size, speed of travel, feed speed, wire diameter, shielding gas, arc voltage, current, nozzle-base distance, etc. These parameters affect by producing spatter in the product, inclusion and other defects, and improper cooling rate, resulting in different microstructures forming along the component's width and a higher residual stress gradient than the ideal. Hence, they need to be monitored carefully during the process [1,3,4].

The products manufactured by WAAM show excellent mechanical properties and are comparable to those produced using traditional manufacturing processes. This is because the wire feed is dense metal and has very little porosity. So, unlike techniques like powder bed fusion, this creates high-quality components that can be readily used [5]. WAAM is extremely cheap as the equipment is existing welding equipment, and the filler metals can also be bought off the shelf, similar to those used in welding. In other processes, creating powder metal is a task that needs to be done carefully as the grain size can influence the strength and appearance of the object, contrary to WAAM, which has no special preparation processes [6]. Since WAAM uses a robotic arm that can move freely, it allows us to 3D print large volumes of metal compared to other DED technologies, which need unique setups. Hence, it can be readily adopted into heavy industries. Powder bed fusion requires vacuum and high-power lasers, both of which make it suitable only for components with smaller build envelopes [7]. WAAM can also be used as an easy fix for damaged components. This will help save components that have just minor damage and

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https://doi.org/10.1016/j.matpr.2022.06.338
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will help save the cost, time, and materials required to manufacture an entirely new component [8].

Even though WAAM has many advantages, as stated above, making it one of the best additive manufacturing methods in large-scale part manufacturing, it has a few limitations due to the varying micromechanical properties attributed to different causes. Most WAAM-manufactured components have simple geometries, but geometrical precision may not be as high as other AM technologies. Hence, this method of additive manufacturing is feasible for less difficult shapes. This dimensional inaccuracy or the limited ability of WAAM to produce finer surface finish, also called the stair-stepping effect, is due to the heat accumulation effects and overheating of the molten pool, making it unstable, decreasing the surface tension. Thus, the accurate control of bead geometry is also affected, leading to variation in microstructure and mechanical properties with the increase of samples' height [9]. Surface Waviness reduces the fatigue life and tensile properties and increases the crack propagation of the as-deposited WAAM part as it acts as a stress raiser [10]. Elastic and plastic deformation is caused by liquid metal deposition on a solid substrate and its constrained shrinkage during cooling. Non-uniform extension caused due to non-uniform constriction results in residual thermal stresses. Tensile residual stresses are the greatest in the longitudinal direction. The magnitude of the residual stresses is determined by the metal's thermal properties and temperature-dependent flow strength. The residual stresses of the material may sometimes be equal to the material's yield stress even at room temperature [11]. Since WAAM involves higher cooling rates and the solidification of melted droplets to form freeform structures, the resulting alloy will have a coarse dendritic microstructure. The innate growth of dendrites and the non-equilibrium thermal history are detrimental to mechanical properties due to the formation of the inter-dendritic Laves phase, which is brittle. The epitaxial dendrite growth due to the WAAM process's directional thermal conduction route helps form very large columnar grains. These grains are heavily textured because they are oriented in a specific direction. As a result, the material exhibits noticeable anisotropy [12]. In other words, we can say that this anisotropy is caused due to the layer-by-layer deposition method used in WAAM processes. Thus, the properties vary from the deposition direction through the thickness. Small pores or cavities are observed mainly in the transitions between layers due to the disintegration of grease, moisture, and other hydrocarbon contaminants and gas entrapment during welding [13]. The size of the pores varies from micropores to coarse pores. Hydrogen porosity is often difficult to control. Because porosity can affect the degradation of mechanical properties of WAAM components when stress is applied in a particular direction, it is one of the common limitations. The studies by Kazanas et al. [14] and L. Yuan et al. [15] show that the fabrication of self-supporting or overhanging structures was a challenge for Wire Arc Additive Manufacturing because of the hump formation and stress formed due to the acute angle. Another limitation is "Delamination" [16]. The deposited layers adjacent to each layer get separated due to improper re-melting of the bottom layer during the deposition process of the top layer.

Although as-fabricated WAAM parts have superior mechanical properties when compared to traditional sand-casting parts, it may be possible to boost their mechanical properties further by mitigation and optimization strategies. This can be mainly done at three different stages. The first is before the deposition process, called pre-process strategies. Pre-process strategies include building strategy optimization, parameters optimization, and clamping. The second is while the deposition process is happening, called online-processing strategies, and the last, after the deposition process, called post-process strategies [17].

In the process of additive manufacturing, one of the important things that we come across and have to decide upon is the scanning path of the machine. It is important because the path plan of the weld bead determines the factors like surface roughness, accuracy of the shape of the structure - which includes the continuity, regularity and thickness, the quality of the distribution of infill like – the control on voids, overlapping of beads and other defects. It also determines to an extent the consumption of the material and the time for printing. Adjusting or optimizing the scanning path can also help to reduce the residual stress and improve the mechanical properties so that the specifications of the structure are met without adjusting the machine or component design [18]. In short, path planning can help in achieving the objectives related to functional properties, topology, and mechanical properties. Conducting trial and error for 3-D printing is a costly and slow process. On the other hand, employing computational techniques such as FEA (Finite element analysis) can be the best alternative to analyze and predict the required result with the least error.

The temperature evolution and cooling rate of an additively manufactured layer depend on the scanning strategy. Thus, 3-dimensional models of different scanning patterns are made to represent the WAAM deposited structure for the numerical simulations in Simufact Welding package by MSC Software Corporation, to compare the residual stresses and distortions which vary with the temperature evolution and cooling rate. The heat source movement and the material deposition are realized in the simulation by dividing the entire process into unit steps in which a specific volume of the material to be deposited is added into the baseplate until the completion of the layer, and to ensure continuity of the layer, the domain of computation is dynamically updated after each step. The results obtained from this simulation is expected to help us in understanding the effects of scanning path planning on the residual stresses and distortion in a layer of the deposited structure and the extent to which it can be optimized for getting a required output.

2. Simulation

2.1. Path planning strategies

A list of studies on different path planning strategies that were conducted is mentioned in the review article by Jiang et al. [18]. Each of the strategies is useful for different and specific applications. In this study, the residual stresses and total distortion of 5 different wire arc additive manufacturing path plans are studied by Finite Element Simulation using Simufact Welding FEA software. Even though the scope of this study is to do a comparative study of the residual stress and distortion that occurs in a single layer due to the temperature evolution, it has been reported that the effect of residual stresses of preceding layers on following layers is insignificant [19]. The path planning strategies that were studied are given in Fig. 1.

2.2. Heat source model and parameters

For simulation of a wire arc additive manufacturing process, it is essential to replace the complex physical processes that generate the heat with a mathematical model of a heat source with appropriate input power distribution to obtain accurate results. As is in all other similar processes, the heat source is used to create the weld-pool by melting the metal using the heat generated from it. In this study, we use the double ellipsoidal modal proposed by Goldak et al. [20], in which a Cartesian system (x', y', z') defines a moving power distribution, separately over two regions. The

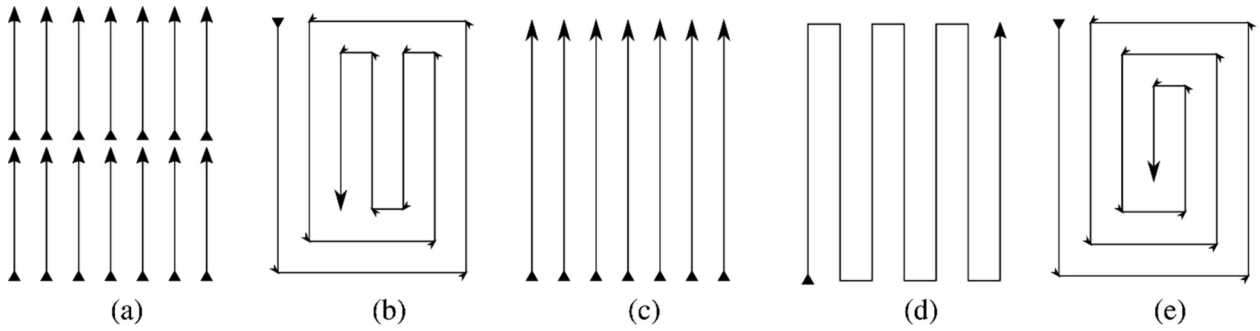


Fig. 1. (a) 2-Section raster pattern; (b) Zig-zag + Spiral pattern; (c) Raster pattern; (d) Zig-zag pattern (e) Spiral pattern.

volumetric power distribution, when $z' > 0$ is given by equation (1), and when $z' < 0$ is given by equation (2).

$$q(x',y',z',t) = f_1 \frac{6\sqrt{3}}{abc_1\pi^{3/2}} \dot{Q} e^{\left\{-3\left[(x'/a)^2+(y'/b)^2+(z'+v(\tau-t)/c_1)^2\right]\right\}} \quad (1)$$

$$q(x',y',z',t) = (2-f_2) \times \frac{6\sqrt{3}}{abc_1\pi^{3/2}} \dot{Q} e^{\left\{-3\left[(x'/a)^2+(y'/b)^2+(z'+v(\tau-t)/c_2)^2\right]\right\}} \quad (2)$$

where, $\dot{Q}$ is the input power, f_1 and f_2 are the weights for the amount of power distributed in front and behind the arc center, satisfying the conditions given by equations (3) and (4), v is the velocity of heat source, and t is the time.

$$f_1 = \frac{2c_1}{c_1+c_2} \quad (3)$$

$$f_1 + f_1 = 2 \quad (4)$$

In the FEM software, the simulation is modeled using the arc welding process. It uses Goldak's double ellipsoidal moving heat source. The parameters used in this experiment are tabulated below: Table 1

The welding (or welder) is represented in Simufact Welding by a robot. A single process may include many robots as needed, and here we used only a single robot for the different cases. Each robot contains its weld paths, trajectory configuration, sequences, and geometry (mesh) for filler. The trajectory files in the.csv extension were made with the help of Python, which contains the coordinates through which the heat source should move each time. The torch angles were adjusted in the robot properties. Every robot must have weld paths, initial temperature settings, and material for filler (if applicable). Optional is weld filler geometry (mesh), which can either be generated by Simufact Welding or imported from external mesh. Here we have designed the filler in Solidworks CAD software, meshed using MSC Apex, and imported it to Simufact Welding. The welding robot moves along the specified trajectory with the heat source at the specified offset distance. New elements are added as the heat source moves along the trajectory line to mimic the original arc welding process (see Figs. 2 and 3(b)).

Table 1
Table of parameters for FE Modelling.

Parameters	Value
a_f (front length)	2 mm
a_r (rear length)	7 mm
b (width)	1.1 mm
d (depth)	5 mm
M (Gaussian parameter)	3
Heat front scaling factor	0.444
Efficiency	0.6
Current	80 Amps
Voltage	10 V
Speed	60 mm/min

2.3. Geometry and meshing

The geometry consists of two parts. One is the weld bead, and the other is the baseplate. The dimensions of the base plate are 175 mm × 141 mm × 10 mm. The geometry of the AM structure is taken as a cuboid of dimension 52.2 mm × 27.5 mm × 5 mm using Solidworks CAD modeling software. This cuboidal geometry is further made of weld beads of rectangular cross-section whose dimensions of the cross-section are 5 mm × 2.2 mm and shape varies with different path planning strategies. Since the scope of this study is to simply find the variation in residual stresses and distortion with the path plan, the shape of the weld bead and the type of material is not going to affect the result much, and hence the study doesn't require intricate details in material type and geometry of the bead. A small space of 0.1 mm is provided in between each bead for the effective transfer of heat (see Fig. 3).

After generating the 3D CAD models of the bead and baseplate, it was imported to MSC Apex for meshing. A 2.5-D mesh of size 0.6 mm, mappable solids, linear element (Hex8) was used for the beads, and a higher size mesh of the same elements was used for the baseplate.

2.4. Assigning material and boundary conditions

The simulation is performed by importing the predefined temperature dependant material data of 16MnCr5-SPM-sw alloy from the Simufact material database for both the base plate and the filler. The ambient temperature of the model is 293.15 K. Gravity (constant) is assumed to be 9.80665 m/s2 along the downward direction of welding. The base plate is constrained by a bearing of exactly the same length and width under it and two circular fixings on either end of the base plate, which is similar to clamps in actual experiments.

3. Results and discussion

The simulation was run for a time, which included the time for completion of welding and then an additional time allowing the structure to cool freely. The effective stress and total distortion were plotted and analyzed. The results are as given below Table 2.

Residual stresses are self-equilibrating stresses that arise out of complex mechanical-thermal-metallurgical interactions like differential heating/cooling etc., in elastic materials during the welding process that often results in distortion during cooling. When the equivalent stress of the as-deposited structure of different path plans was plotted using the contour plot (Fig. 4), it was found that

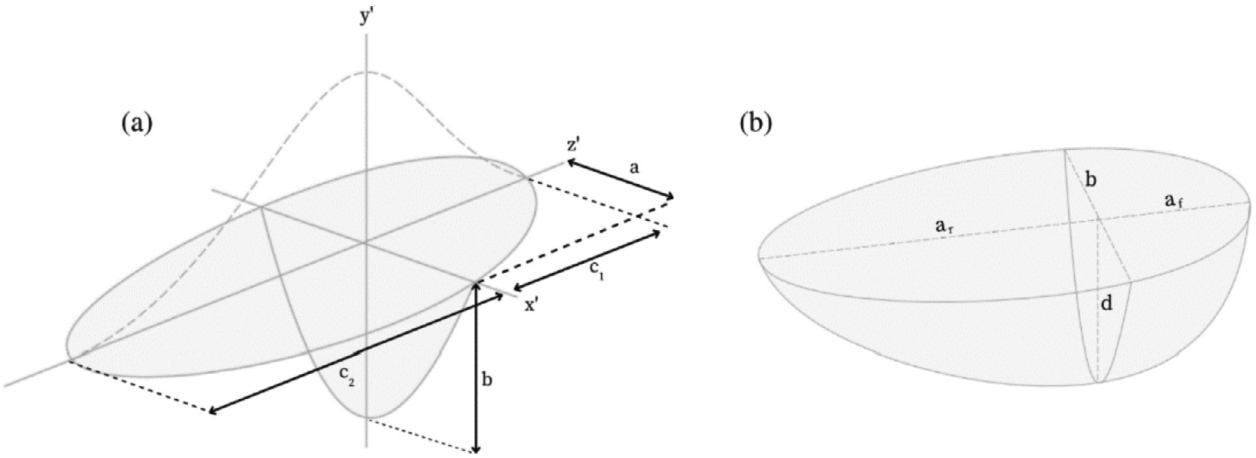


Fig. 2. (a) Heat input model proposed by Goldak et al.; (b) heat source model used in the simulation.

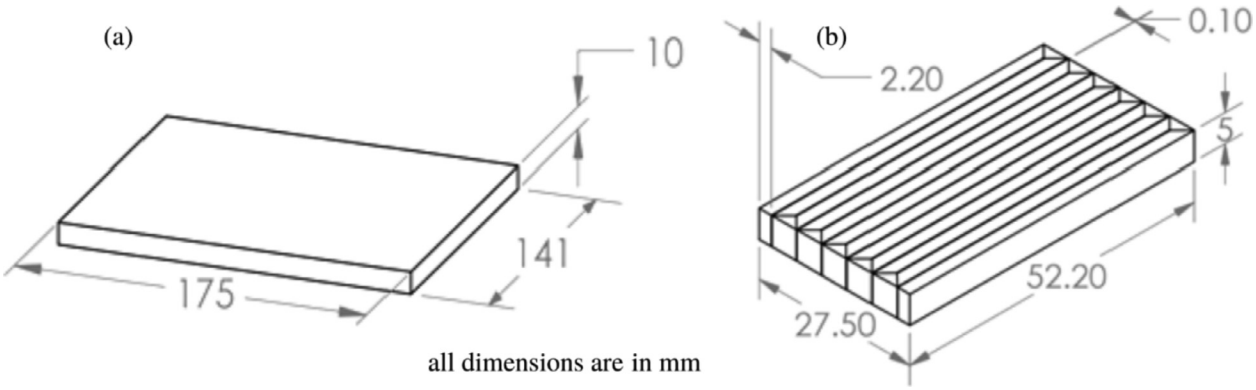


Fig. 3. Geometry of (a) baseplate; (b) weld geometry (AM structure).

Table 2
Summary of the results.

Path planning strategy	Maximum effective stress (MPa)	Total distortion (mm)
Spiral	385.80	0.66
Zigzag	388.03	0.86
Zigzag + Spiral (Contour offset)	389.28	0.71
Raster	385.66	1.03
2-Section Raster	411.46	0.74

the least residual stress occurred in the spiral and raster scanning path, and the maximum was obtained in 2-section raster path. The contour plot of total distortion shows that the total distortion was maximum in raster and minimum in spiral strategy. It is also worth noting that the distortion in the case of each of the strategies are mainly taking place in the longitudinal direction of the bead (see Fig 5)

The minimum residual stress is observed when the spiral and raster scanning paths were used, and the maximum was obtained in the 2-section raster path. However, the distortion was maximum in raster and least in spiral strategy. It could be inferred from this that whichever path has a bead welded next to it after being given enough time for cooling will have the least residual stress. This is because, the temperature difference created results in faster cooling of the bead deposited adjacent to it. Here, the spiral path has an adjacent bead welded after one complete rotation along the contour of the additively manufactured geometry. In the case of raster strategy, the adjacent layer of bead is built in the same order after one complete layer is done. Here we can see that, in both these strategies, the beads are getting enough time to get cooled and hence less residual stress due to uniform compression as a result of uniform cooling. While, in the 2-section raster path, the adjacent bead is deposited before giving enough cooling time to the previous layer, thus giving the structure a temperature gradient, which leads to non-uniform cooling, and further non-uniform compression in the geometry, leading to increased residual stress.

The distortion will be maximum when the bead can undergo distortion in different directions. That is, when a bead has more surfaces open to the external surface, it is free to distort. As we noted the distortion mainly along the longitudinal direction, it can be inferred that the beads are freer to distort along that direction. This may be attributed to the temperature distribution along a bead when compared to the surroundings, and the hindrance caused by surrounding beads for the lateral expansion. In the case of a single-section raster path, each of the beads has open surfaces at both ends and hence more distortion. In the case of the spiral pattern, which has the least open cross-section of beads, it has the least distortion for the same reason. However, it is obvious for one to have a doubt on why the spiral + offset pattern doesn't show the same effect. This can be explained by considering the zig-zag beads that happen to be inside the pattern. The expansion of these zig-zag beads along the longitudinal direction might have pushed the spiral beads outside to increase the resultant distortion. However, this reasoning might not work well with all similar cases, as it varies with different geometries.

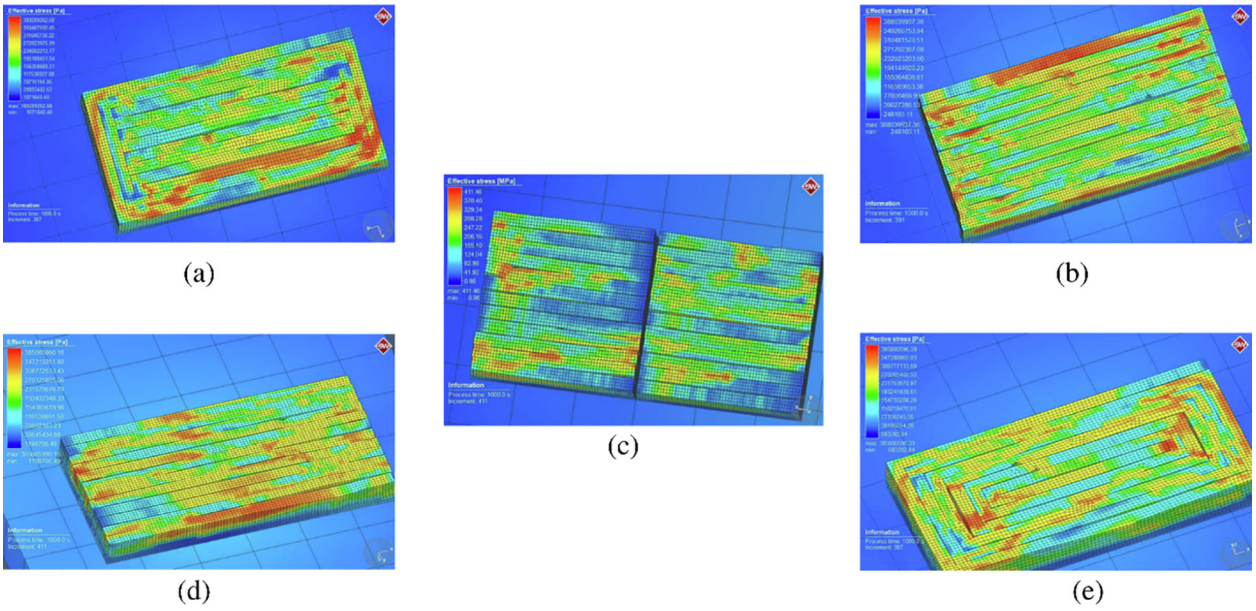


Fig. 4. Effective stress contour of (a) zig-zag + spiral strategy; (b) zig-zag strategy; (c) 2 section raster strategy; (d) raster strategy; (e) spiral strategy.

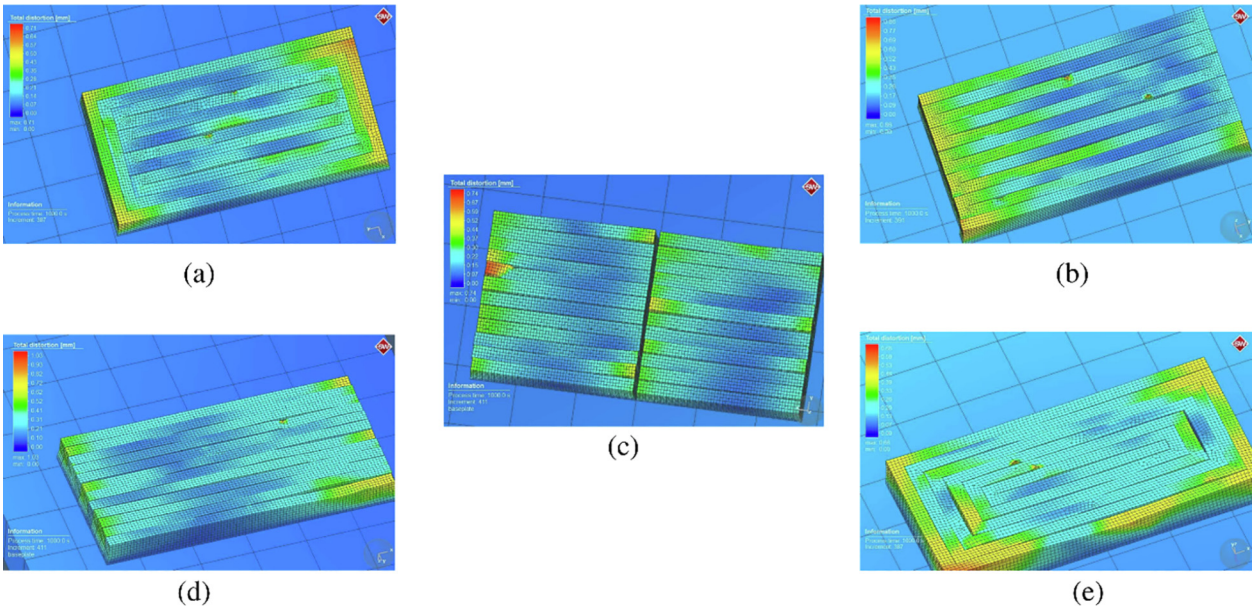


Fig. 5. Total distortion contour of (a) zig-zag + spiral strategy; (b) zig-zag strategy; (c) 2 section raster strategy; (d) raster strategy; (e) spiral strategy.

4. Conclusion

Path planning plays a crucial role in determining the quality, property, and many other factors of an as-deposited structure. Very few research works were only reported to be done on the effect of path planning. CAD models of different strategies of a layer of weld bead and the baseplate were made. It was then meshed and FEM simulation of WAAM was done as unit steps with the moving heat source distribution and the addition of new elements along with it. The result of the finite element study is reported in the above section. It was seen that the least residual stresses were in spiral and raster paths whereas, the highest was in 2-section raster strategy and the distortion was least in spiral strategy and highest in raster strategy. It was concluded that the structure that will have the least non-uniform cooling will have the least residual stress and because of the temperature profile along the bead, the paths in which fewer longitudinal beads are exposed to the edges will tend to distort the least.

The geometry used to study the path planning strategies is generic to a particular case wherein bulk geometries are being constructed. But the result yielded could comparatively change if the geometry of the build is comparatively different. There is a need for standard geometry in order to evaluate the properties of an additively manufactured specimen. Only five types of path planning strategies were studied. Including more path planning strategies can help in reaching a proper conclusion on the factors that affect the properties and their variation with them. A conventional model was used in the study. Better results can be achieved if we incorporate the multi-physics involved in the welding process.

CRediT authorship contribution statement

M.S. Amal: Methodology, Software, Writing – original draft, Visualization, Investigation. **C.T. Justus Panicker:** Conceptualization, Writing – review & editing. **V. Senthilkumar:** Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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